Mid Term Odd Semester Examination, Session 2019-20
M. Tech. Part I (Ist Semester) & Ph. D. Course Work
Mining Engineering
Subject: Geo-informatics, Subject Code: MN542

Time: 2 hours **Full Marks: 30**

Note: ***1. Answer all questions.***
2. Make all suitable assumptions wherever necessary.

1. Do as directed: **[1 X 5]**

(a) A digital image having a radiometric resolution of 6 bit, _____ is the maximum value and _____ is the minimum value of the DN could be represented by that image.

(b) If the IFOV for all pixels of a scanner stays constant it means that _____ resolution will vary from the image centre to the _____ edge.

(c) State the name of any two sensors with its resolution.

(d) Differentiate between multispectral and hyperspectral sensors?

(e) If all visible wavelengths are reflected from an object, it will appear _____, while an object absorbing all visible wavelengths will appear _____.

2. **(a)** What type wavelength and sensor would be the best to monitor the general health of all vegetation cover over an area for several months, and why? **[2]**

(b) A company wants to start a surface mine in a mountainous region. Limiting geometric distortions as much as possible, would you choose a satellite-based or aircraft-based scanning system? Explain in terms of imaging geometry. **[3]**

3. What is DGPS? Explain the working principle and its various methods of measurement using examples. **[5]**

4. **(a)** Why are most remote sensing systems avoided detecting and recording wavelengths in the ultraviolet and blue portions of the spectrum? **[2]**

(b) Discuss the trade-off between resolutions? Considering an example, how is it significant for your work? **[3]**

5. **(a)** How does the EMR interact with; (i) The Earth (ii) The Atmosphere. **[3]**

(b) What do you understand by the term atmospheric window? Explain. **[2]**

6. **(a)** If the spectral range of the 400 channels of the CASI (Compact Airborne Spectrographic Imager) is exactly 0.40 μm to 0.90 μm and each band covers a wavelength of 1.25 nm will there be any overlap between the bands? **[2]**

(b) Differentiate between Supervised and Unsupervised classification with suitable examples. **[3]**

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